

ADVANCED · FULL-SPECTRUM · AI-NATIVE

APP DEVELOPMENT ENGINEERING

A 40-session advanced program spanning cross-platform mobile, cloud-native backends, DevOps automation, on-device AI and production-grade security — built with six industry capstones.

Flutter

React Native

Kotlin / Swift

Node · GraphQL

CI/CD

On-device AI

MOBILE / WEB / CLOUD / DEVOPS / AI — ONE CONTINUOUS TRACK

6 CAPSTONES · DEMO DAY

40

SESSIONS

06

CAPSTONES

150+

HIRING PARTNERS

24/7

MENTOR SUPPORT

BUILT FOR THE NEXT DECADE

Edufyi Tech Solutions runs industry-embedded virtual internships and training tracks for college students. Every learner ships six real-time and capstone products under mentors from MNCs and venture-backed startups.

1

MENTOR-LED

Live architecture reviews and 1:1 code critique from senior engineers.

2

PRODUCTION-FIRST

You ship to real stores and real clouds, not to localhost.

3

OUTCOME-BACKED

Placement assistance, mocks, GDs and performance stipends.

WHY CHOOSE US



Self-paced + cohort tracks



Industry project playground



Resume & aptitude grooming



Mentors from MNCs & founders



100% placement assistance



Exclusive R&D content



Unlimited mocks & GDs



24x7 dedicated support



Wipro DICE ID certification



Internships with 150+ firms



1:1 counselling sessions



Microsoft · Cisco · IBM · Adobe



Certified bootcamps & webinars



Lifetime personalised LMS



Recorded lectures anytime



Performance-based stipend

WHAT MODERN APP ENGINEERING REALLY IS

App development today is a full product lifecycle: designing interfaces, engineering cross-platform clients, wiring cloud-native APIs, automating delivery pipelines, instrumenting telemetry and hardening security — then iterating on real usage data.

EXPERIENCE LAYER

Design systems · Motion · Accessibility · Offline-first UX

01

CLIENT LAYER

Flutter · React Native · Kotlin · SwiftUI · State machines

02

API LAYER

REST · GraphQL · gRPC · WebSockets · Edge functions

03

DATA LAYER

PostgreSQL · MongoDB · Redis · Vector stores · Sync engines

04

PLATFORM LAYER

Docker · CI/CD · Observability · Cost & release governance

05

MARKET SCOPE

USD 6.9B → 14.3B

APP DEVELOPMENT MARKET, 2021 → 2030



PHASE 01 — CLIENT ENGINEERING

SESSIONS 1 – 18 • WEB + APP FOUNDATIONS

S 01-03

FOUNDATIONS

- Modern web & app landscape, SDLC
- Git, branching and code review flow
- HTML5 semantics & accessibility

S 04-06

CSS ENGINE

- Flexbox, Grid, container queries
- Design tokens & theming
- Responsive and fluid typography

S 07-09

JAVASCRIPT CORE

- ES2023 syntax, scope, closures
- Async, promises, event loop
- Modules, bundlers, tooling

S 10-12

TYPESCRIPT

- Types, generics, narrowing
- Interfaces & utility types
- Typed API contracts

S 13-15

REACT MASTERY

- Hooks, context, custom hooks
- Rendering & memo strategy
- Routing and code splitting

S 16-18

STATE & DATA

- Query caching & mutations
- Global state machines
- Optimistic UI and offline cache

PHASE OUTCOMES

- Build accessible, responsive interfaces with a reusable design system
- Write typed, testable JavaScript / TypeScript at production quality
- Architect React apps with predictable state and fast rendering

LAB DELIVERABLE

Portfolio-grade component library
 Typed dashboard with live data
 Performance audit report

PHASE 02 — PLATFORM & BACKEND

SESSIONS 19 - 32 · MOBILE CLIENTS + CLOUD APIS

S 19-21

CROSS-PLATFORM

- Flutter widgets & Dart basics
- React Native + Expo pipeline
- Native modules and bridges

S 22-23

NATIVE TRACK

- Kotlin & Jetpack Compose
- SwiftUI fundamentals
- Platform permissions & lifecycle

S 24-25

MOBILE UX

- Navigation & deep linking
- Gestures, motion, haptics
- Push notifications

S 26-28

NODE & EXPRESS

- HTTP, middleware, routing
- Auth, sessions and cookies
- File uploads & streaming

S 29-30

DATABASES

- MongoDB & PostgreSQL modelling
- Indexes, aggregation, migrations
- Redis caching patterns

S 31-32

API DESIGN

- REST versioning & pagination
- GraphQL schemas & resolvers
- WebSockets & realtime sync

PHASE OUTCOMES

- Ship one Flutter and one React Native app to real devices
- Design secure REST and GraphQL APIs with proper data modelling
- Implement realtime sync, notifications and offline behaviour

LAB DELIVERABLE

Cross-platform app build
Authenticated API service
Realtime chat module

PHASE 03 — SHIP TO PRODUCTION

SESSIONS 33 - 40 · CLOUD, DEVOPS, AI, SECURITY

S 33-34

CLOUD DEPLOY

- Containers & image builds
- Serverless and edge functions
- Environment & secret management

S 35-36

CI / CD

- Automated test pipelines
- Fastlane store releases
- Feature flags & rollbacks

S 37

SECURITY

- OAuth2, OIDC, JWT rotation
- Secure storage & encryption
- OWASP MASVS review

S 38

AI INTEGRATION

- LLM APIs & streaming responses
- Embeddings and RAG retrieval
- On-device inference

S 39

OBSERVABILITY

- Crash reporting & tracing
- Product analytics funnels
- Performance budgets

S 40

SHIP & SCALE

- Store listing and ASO
- Monetisation & subscriptions
- Capstone demo day

PHASE OUTCOMES

- Containerise and deploy services with automated CI/CD pipelines
- Harden apps against the OWASP mobile top-10 threat set
- Integrate LLM and on-device inference features responsibly

LAB DELIVERABLE

Automated release pipeline
Security review report
AI-powered app feature

THE STACK YOU WILL MASTER

LANGUAGES

TypeScript

Dart

Kotlin

Swift

JavaScript

SQL

FRAMEWORKS

React

React Native

Flutter

Node.js

Express

Next-gen SSR

DATA

MongoDB

PostgreSQL

Redis

Prisma

Firebase

Vector DB

DEVOPS

Git

Docker

GitHub Actions

Fastlane

Sentry

Cloud CLI

AI & ML

LLM APIs

Embeddings

TF Lite

Core ML

Prompt design

RAG

DESIGN

Figma

Design tokens

Motion

Accessibility

Prototyping

Analytics

PROGRAM ROADMAP

WEEK 01-04



Foundations

WEEK 05-09



React + TS

WEEK 10-14



Mobile builds

WEEK 15-18



APIs & data

WEEK 19-22



Cloud & CI/CD

WEEK 23-26



AI + Capstones

EVERY PHASE ENDS WITH A GRADED BUILD, A CODE REVIEW AND A MENTOR DEBRIEF

ADVANCED SPECIALISATIONS

Elective deep-dives layered on top of the core track.

ON-DEVICE AI

- TensorFlow Lite / Core ML
- LLM APIs & streaming UX
- RAG + vector search
- Prompt & token cost control

CLOUD NATIVE

- Docker & container registries
- Serverless & edge runtimes
- Autoscaling & queues
- Infra as code

DEVOPS & RELEASE

- GitHub Actions pipelines
- Fastlane store automation
- Staged & canary rollout
- Crash + APM telemetry

APP SECURITY

- OAuth2 / OIDC & JWT rotation
- Secure storage & keychain
- OWASP MASVS checklist
- Pen-test & obfuscation

PERFORMANCE

- Render & frame budgets
- Bundle and cold-start tuning
- Memory & battery profiling
- Offline sync + caching

PRODUCT CRAFT

- Design systems & tokens
- A/B tests & analytics
- Monetisation & IAP
- ASO and growth loops

SIX CAPSTONE BUILDS

CAPSTONE 01

Realtime chat & video app

CAPSTONE 02

Fintech wallet with KYC flow

CAPSTONE 03

AI health companion

CAPSTONE 04

Hyperlocal delivery platform

CAPSTONE 05

Learning app with offline sync

CAPSTONE 06

IoT device control dashboard

CAREERS & ANALYTICS

● **Mobile App Engineer** INR 4.5 - 12 LPA

● **Flutter / RN Specialist** INR 5 - 14 LPA

● **DevOps Associate** INR 5 - 12 LPA

● **Game Developer** INR 4 - 11 LPA

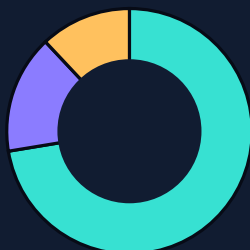
● **Full-Stack Developer** INR 4 - 11 LPA

● **Backend / API Engineer** INR 5 - 13 LPA

● **UX Engineer** INR 4 - 10 LPA

● **AI App Developer** INR 6 - 16 LPA

LEARNER BACKGROUND

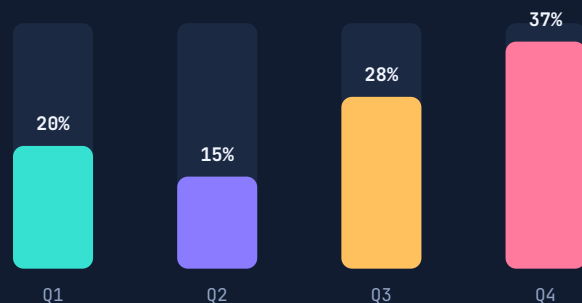


■ UG Management 72.4%

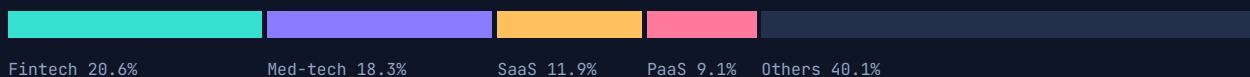
■ UG B.Tech/BE 15.6%

■ Others 12.0%

QUARTERLY PLACEMENTS



HIRING SECTORS



ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



INTERNSHIP CERTIFICATE
Accredited Certification



INDUSTRIAL TRAINING
Accredited Certification



PROJECT COMPLETION
Accredited Certification

INTERNSHIP

Verified internship completion credential

TRAINING

Industrial training hours certified

PROJECT

Capstone project completion proof

GLOBAL MNC CERTIFICATIONS.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



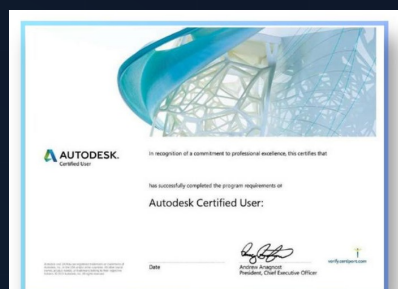
CISCO
Cybersecurity CCST



MICROSOFT
Azure Fundamentals



IBM
Internet of Things



AUTODESK
Certified User



ESB
Entrepreneurship

AUTHORISED PARTNERS.

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



ADOBE



MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

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Fundamentals



Cisco Cybersecurity CCST



IBM Internet of Things



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Professional



Autodesk Certified User



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